Fish Abundance Over Time

Exploring common manipulations, working with dates and time, arrange(), and visualizing missing data

Set Up

```
# loading in packages
library(tidyverse) # general use
library(janitor) # cleaning data
library(here) # data organization
library(lubridate) # date and time
library(naniar)

# reading in data set
fish <- read.csv(here("data",
                      "annual_all_species_fish.csv"),
                 # changing -99999 values to be na
                 na = "-99999")
```

Cleaning Data

```
# creating new object, fish_clean
fish_clean <- fish |>
  # cleaning the names
  clean_names() |>
  filter(group == "FISH") |>
  # renaming the date column to original_date
  rename(original_date = date) |>
  drop_na(dry_gm2) |>
  # using mutate and lubridate to convert original_date
  # into new date column that can be used to create time series
```

```
  mutate(
    date = ymd(original_date),
    year_month = floor_date(date, unit = "month")) |>
  # relocating columns to to left to be easier to look at
  relocate(year_month, dry_gm2, common_name, site)
```

Common Manipulations

Rename (function in dplyr)

The rename() function allows you to change the names of values for new ones.

In this data set: rename() was used to change the date column to be named original_date. This was useful so a different date column could be created through mutate, and date was the preferred column name.

Relocate (function in dplyr)

The relocate function allows for columns positioning in a data set to be changed. This was useful to look at specific columns together by utilizing relocate which automatically pulls the specified columns to the left.

Working with Dates and Time

Utilizing lubridate package

Lubridate is a package that is used to help make working with dates and times easier.

In this data set:

- ymd() was used to turn the original dates into a version of dates that R can better recognize and put it in a year day month format
- floor_date() takes the new "date" column and rounds each date down to the first of the month. This allows for the mean dry mass by month to later to explored in the time series.

Other Functions Used

- clean_names(): used to simplify the column names from the original data file
- drop_na(): used to drop missing values from data to better prepare for summarize function use

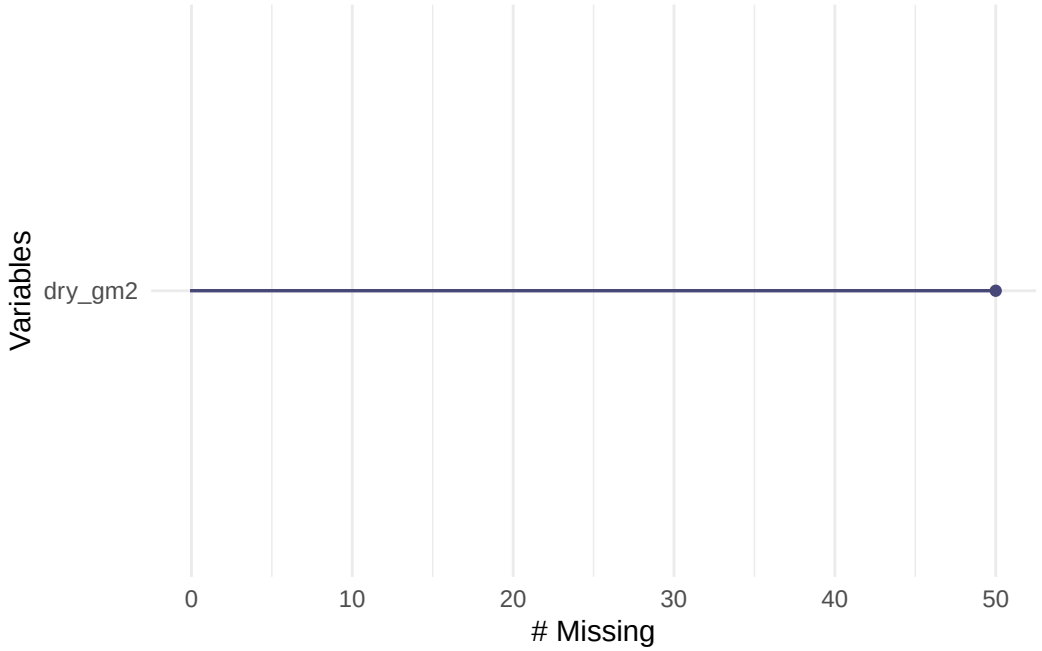
Exploring Data

```
# creating top_species_table
top_species_table <- fish_clean |>
  # grouping by species
  group_by(common_name) |>
  # summarize to make one row per species
  # sum to get the total fish count of each species
  summarize(total_dry_gm2 = sum(dry_gm2)) |>
  # ungrouping the data
  ungroup() |>
  # ranking from most to least abundant
  arrange(desc(total_dry_gm2)) |>
  # selecting top 5 species
  slice_head(n = 5)

# creating top_species_vector
top_species_vector <- top_species_table |>
  # using pull to extract column as a vector
  pull(common_name)
```

Visualizing Missing Data

```
# looking at fish_clean
fish |>
  clean_names() |>
  # filter for fish
  filter(group == "FISH") |>
  # filter for top species
  filter(common_name %in% top_species_vector) |>
  # selecting for dry mass
  select(dry_gm2) |>
  # using gg_miss_var to check for "na" data
  gg_miss_var()
```

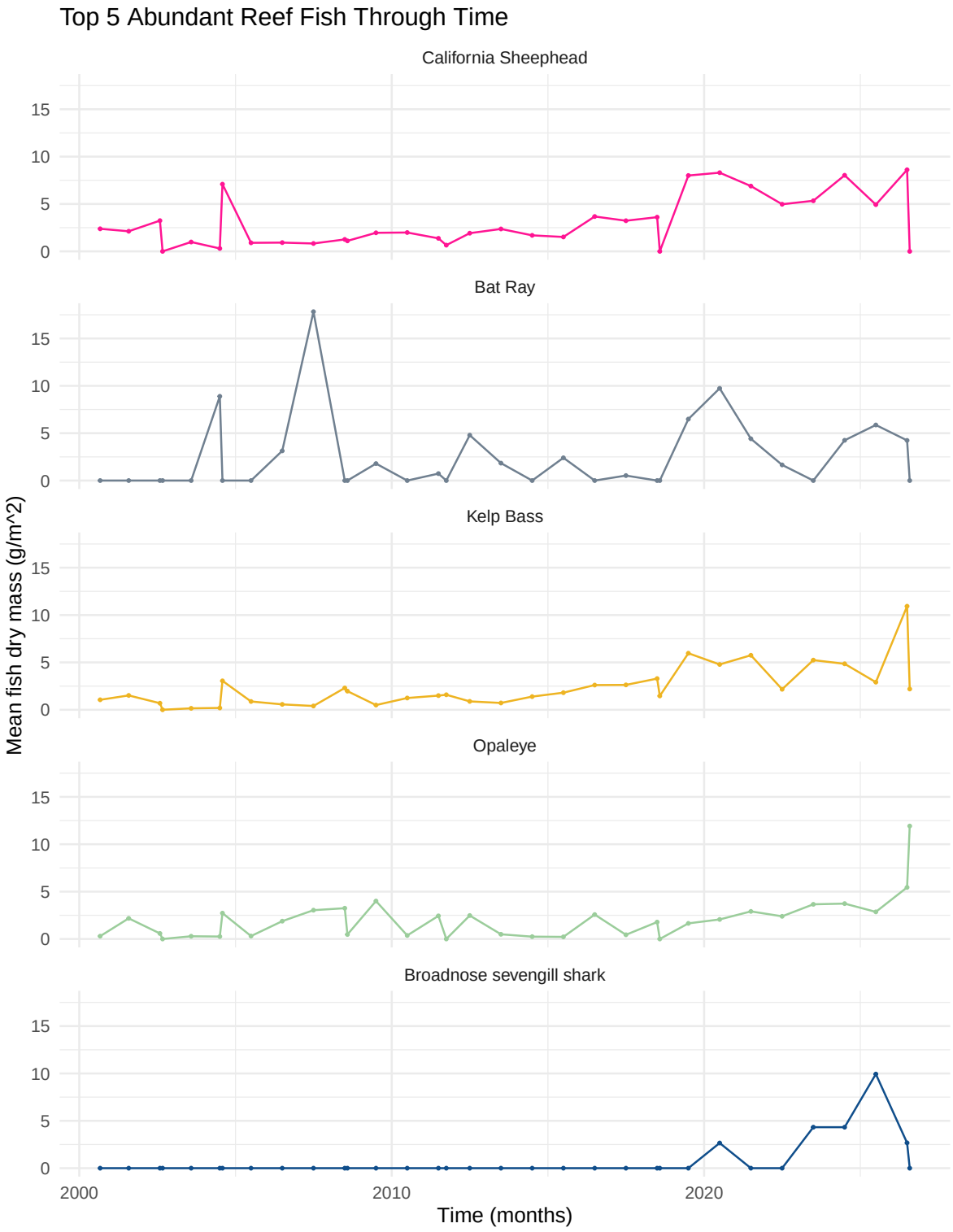


Preparing Data for Timeseries

```
# creating new object fish_ts
fish_ts <- fish_clean |>
  # filtering for the top_species_vector in common_name
  filter(common_name %in% top_species_vector) |>
  # using fct_relevel to change order within common_name
  mutate(common_name = fct_relevel(
    common_name,
    top_species_vector)) |>
  # grouping data by date and species
  group_by(year_month,
           common_name) |>
  # summarize to make one row per month
  # sum to get the mean dry mass for each species
  summarize(mean_dry_gm2 = mean(dry_gm2)) |>
  # ungrouping the data
  ungroup() |>
  # arranging by most abundant species and by date
  arrange(common_name, year_month)
```

Top 5 Fish Species Timeseries

```
# base plot
ggplot(fish_ts,
       # x-axis: time
       aes(x = year_month,
           # y-axis: abundance
           y = mean_dry_gm2,
           # coloring by species
           color = common_name)) +
  # first layer: geom_line
  geom_line() +
  # second layer: geom_point
  geom_point(size = 0.5) +
  # faceting by species
  facet_wrap(~ common_name,
             # putting into one column
             ncol = 1) +
  labs(
    # adding a graph title
    title = "Top 5 Abundant Reef Fish Through Time",
    # naming x axis Time
    x = "Time (months)",
    # naming y axis Total fish count
    y = "Mean fish dry mass (g/m^2)") +
  # customizing colors
  scale_color_manual(values = c(
    "California Sheephead" = "deeppink1",
    "Bat Ray" = "slategray",
    "Kelp Bass" = "goldenrod2",
    "Opaleye" = "darkseagreen3",
    "Broadnose sevengill shark" = "dodgerblue4")) +
  # adding custom theme
  theme_minimal() +
  # removing legend
  theme(legend.position = "NONE")
```


Top 5 Abundant Reef Fish Through Time
California Sheephead
Bat Ray
Kelp Bass
Opaleye
Broadnose sevengill shark
Mean fish dry mass (g/m^2)
0
5
10
15
2000
2010
2020
Time (months)